



X4Pro - universal, fully relational EXFOR database

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This page presents progress in development X4Pro database and tools ([download](#)).

Concept.

X4Pro extends EXFOR-Relational database with experimental data points in original and computational form and additional information for data renormalization.

The X4Pro database provides:

- access to all EXFOR meta-data and numerical data using only SQL commands, i.e. no need in original EXFOR, EXFOR parser and converter
- data for renormalization: old and new monitor cross sections, old and new decay data (gamma lines intensity of measured reaction and monitor product)
- experts' instructions for EXFOR data modifications
- EXFOR Subentries and Datasets in X5-json format (since 2023-03-30)
- evaluators' flags and scores (since 2025-01-10)

Tasks of the project:

1. Desing of database schema
2. Work out data and programming needs on typical examples
3. Develop new concept of EXFOR data corrections and define workflow
4. Envent/find out possible technical solutions for existing tasks, study problems, implement solutions
5. Prepare for test/demo examples for end-user, confirm working tools on platfroms/enviromnent:
 - a) X4Pro: local SQLite database and remote MariaDB database
 - b) python on Windows, Linux, MacOS (plot with Plotly and Mathplotlib)
 - c) gfortran, gcc on Windows, Linux, MacOS
 - d) retrieve ENDF data from Web server and plot together with EXFOR data
 - e) populate NoSQL database (CouchDB) using X5-json from X4Pro as data source

Distribution (see [readme](#), [license](#), [copyright](#)), ver.2026-09-29:

- X4Pro/SQLite database, ver.2026-09-29
- Python and Fortran demo-programs, ver.2026-09-29
- Bash-scripts running demo-programs (CLI [\[1\]](#) and TUI [\[2\]\[3\]](#)), database statistics, ver.2026-09-29
- consists of 6 parts:

#	Dir	Description	SF:MF	Notes	Demo plot/output [code][modules]	Comment
Part-1. Retrieve data from local EXFOR database Python. Retrieve experimental data from X4Pro database and plot data using Plotly amd Matplotlib						
0.	part1-0-sig	Cross sections	SIG	Al-27(n,a):CS(Ei)	Al27na-1.html/json [code] Al27na-2.html/json [code]	Code: single .py file for copy/paste 2) data grouped Reaction-codes
1.	part1-1-sig	Cross sections	SIG	Al-27(n,a):CS(Ei)	Al27na-1.html/json [code][rx4db] Al27na-2.html/json [code]	Coding: modular (several .py files) 2) data grouped Reaction-codes
2.	part1-2-da	Angular distributions	DA	O-16(a,e):DA(Ei)	da0an.html/pdf/json [code][rx5da]	Plotly/matplotlib
3.	part1-3-dap	Angular distributions (partial)	PAR,DA	Li-6(he3,p):DA(θ)	dap0ei.html/pdf/json [code][rx4dap_ei]	Plotly/matplotlib
4.	part1-4-de	Emission spectra	DE	Th-232(n,xn):DE(Eo)	de0eo.html/pdf/json [code][rx5de]	Plotly/matplotlib
5.	part1-5-dae	Double differential cross sections	DA/DE	F-19(n,xn):DAE(Eo)	dae0eo.html/pdf/json [code][rx4dae_e2]	Plotly/matplotlib
6.	part1-6-fy	Fission yeild (mass distribution)	FY	U-238(n,f):FY(A) Ei14MeV U-235(n,f):hist2(A,TKE)	U238nf-fy.html/pdf/json [code][rx4fya] me0x.html [code] me0mpl3d.pdf [code] me0mpl2d2.pdf [code]	1) Plotly/matplotlib 2) Plotly: Contour-2D 3) Matplotlib: plot_surface-3D 4) Matplotlib-2d, 2 subplots
7.	part1-7-covar	Covariance data stored in EXFOR	SIG	1) ²⁴¹ Am(n,2n) correlations ERR-T,MONIT-ERR,ERR-5 2) Reaction-reaction correlations	covar1.tto.html/pdf/json [code] covar2.tto.html/pdf/json [code]	1) Matplotlib-2d, 3 subplots 2) Matplotlib-2d, 1 subplot
8.	part1-8-reac 2026	Any type of EXFOR data	SIG, DA DE, DAE CSP, DAP TKE, ETA NUDAT FY ELEM/ MASS FY NU/DE	1) Partial cross section CSP 2) Temperatire dependent CST 3) Total kinetic energy TKE 4) Neutron yield ETA 5) Neutron yield NUBAR 6) DATA and DATA-MIN 7) Search by author 8) Production cross section, where product coded as SF4 or in DATA (ELEM/MASS) 9) PFNS prompt-fission neutron energy spectra -- etc.	1) CSP: sig1p.html/png/json , tt-out 2) CST: cst.html/png/json , tt-out 3) TKE: tke1.html/png/json , tt-out 4) ETA: eta1.html/png/json , tt-out 5) NUBAR: nu1.html/png/json , tt-out 6) DATA-MIN: si28nn.html/png/json , tto 7) Author1: Kokkoris.html/png/json , tto 8) Prod: sig1prod.html/png/json , tto 9) PFNS: pfns1.html/png/json , tto PFNS (SF): pfns4.html/png/json , tto - [reac1.py] main code - [reac1help.py] display help-text - [reac1subr.py] retrieval subroutines	1) parameters: reaction-code, x-variable, filters 2) descriptions, x-variables for grouping are taken from the database ---under development---
	./	Common modules:			1) [dbConn.py] database connectivity with SQLite 2) [sqlsubr.py] implementation of SQL operations 3) [x4out.py] output datasets prepared for plotting (now: to JSON) 4) [exfor2plot-plotly.py] EXFOR plotting with Plotly 5) [exfor2plot-matpl.py] EXFOR plotting with Matplotlib	

Fortran. Retrieve X4Pro data from Fortan program via C subroutines						
8.	part1-fortran/	0) Simple example Fortran executing SQL 1) Double differential cross sections 2) Cross sections 3) Cross sections	-- DA/DE SIG SIG	First 8 Entries from F-Area F-19(n,xn):DAE(Eo) Mn-55(n,a):CS(Ei) Mn-55(n,a):CS(Ei) →C4	f1.tto.html [code] dae1e2.tto.html [code] sig1.tto.html [code] sig1toc4.tto, C4 [code]	No plots, output text only
9.	part1-fortran/ See also implemen- tation in Python: part3-7-legrs2da	4) Retrieve LEG/RS and SIG from the same Entry; calculate missing Legendre coefficient L[0]; calculate angular distributions	LEG/RS SIG	Cu-0(n,el):DA,,LEG/RS(Ei) SIG→L[0] L[0..n],θ→DA(θ)	legrs2da.tto.html, C4, ZVD [code] terminal:output	FIND: SIG from <Entry, Reaction, Ei> CALC: SIG/4π → L[0]; CALC(Ei): L[0..n],θ → DA(Ei,θ)
	part1-fortran/	Common subroutines:			c1) [sql1sub.c]: fortran-to-sqlite3 adapter (by V.Zerkin) c2) [sqlite3.c][sqlite3.h] sqlite3 amalgamation (by SQLite team)	
Part-2. Retrieve local EXFOR and remote ENDF data						
Python. Retrieve and plot experimental (local EXFOR) and evaluated data (Web ENDF-API)						
1.	part2-1-sig1	Cross sections	SIG	Al-27(n,a):CS(Ei)	Al27na.html/pdf/json [code][rx4db] Al27na2.html/json	1) using Plotly and Matplotlib 2) data grouped Reaction-codes
2.	part2-2-da1an	Angular distributions (Ei:fixed)	DA	Mn-55(a,el):DA(θ)	da1an.html/pdf/json [code][rx5da]	DA + MF4/MF34#new
3.	part2-3-da1ei	Angular distributions (Angle:fixed)	DA	C-12(a,el):DA(Ei)	da1ei.html/pdf/json [code][rx4da_ei]	Plotly/matplotlib
4.	part2-4-de1eo	Emission spectra	DE	Th-232(n,xn):DE(Eo)	de1eo.html/pdf/json [code][rx5de]	Plotly/matplotlib
5.	part2-5-dae1eo	Double differntial cross sections	DA/DE	F-19(n,xn):DAE(Eo)	dae1eo.html/pdf/json [code][rx4dae_e2]	Plotly/matplotlib
6.	part2-6-fy	Fission yeild (mass distribution)	FY	U-238(n,f):FY(A) Ei14MeV	U238nf-fy.html/pdf/json [code][rx4fya][rweb1fy]	Plotly/matplotlib
7.	part2-7-fye 2024	Fission yeild (energy distribution)	FY	U-238(n,f):FY(E)	u235-nf-cd115g-cumFY.htm/pdf/json [code][rx4fye][e4subr]	Plotly/matplotlib
8.	part2-8-sig1par 2026	Partial cross section	CSP	Li-7(n,inl)Li-7,par,sig Li-7(n,n'):MF3.MT51	data:Li7ninl.html/png/json, tt-out - [sig1par.py] main code - [rx4sig1par.py] retrieval subroutines	---in progress---
9.	part2-9-nubar 2026	Neutron yield (nu-bar)	NU	U-235(n,f),[pr dl],nu MT452/455/456:MF1+MF31	PR-data:U235nf.html/png/json,ex,ev,tto DL-data:U235nf.html/png/json,ex,ev,tto TOT- data:U235nf.html/png/json,ex,ev,tto - [nubar.py] main code - [rx4nubar.py] retrieval subroutines	---in progress---
10.	part2-A-eta 2026	Eta: average neutron yield per nonelastic event	ETA	U-235(n,abs)(n,non),,eta MT452:MF1+MF31 MT18:MF3+MF33 MT102:MF3+MF33	U235-eta.html/png/json,ex,ev,tto U233-eta.html/png/json,ex,ev,tto Pu239-eta.html/png/json,ex,ev,tto - [eta.py] main code - [rx4eta.py] retrieval subroutines	---in progress---
11.	part2-B-pfns 2026	PFNS - energy spectrum of prompt fission neutrons	NU/DE	*(N,F),PR,NU/DE* MF5+MT18,...	U-233(n:thermal,f) html/png/json,ex,ev,tto U-235(n:th,f) html/png/json,ex,ev,tto U-235(n:0.5MeV,f)html/png/json,ex,ev,tto U-238(n:14.7MeV,f) html/png/json,ex,ev,tto Pu-239(n:th,f) html/png/json,ex,ev,tto Pu-239(n:1.45MeV,f) html/png/json,ex,ev,tto Pu-239(n:14MeV,f) html/png/json,ex,ev,tto - [reac2pfns.py] main code - [reac2subr.py] retrieval subroutines - [reac2proc.py] convert to ratios - [fort2pfns.py] calculations	---in progress---
		Additional common modules:			6) [websubr.py] GET: download text from NDS web-site 7) [rweb11.py] ENDF Web retrieval (SIG) 8) [rweb12.py] ENDF Web retrieval (DA, DE, DAE) 9) [endf2plot-plotly.py] ENDF plotting with Plotly 10) [endf2plot-matpl.py] ENDF plotting with Matplotlib	
Part-3. EXFOR data corrections and recalculations (for experts)						
Python. EXFOR data corrections (automatic*, users, experts*) and calculations + plot with ENDF data *Note. All input for corrections (monitor CS data, decay data, experts' operations) are stored in the database.						
1.	part3-1-auto1	Automatically renormalize CS using old monitor CS and new standards	SIG	Mn-55(n,a):CS(Ei) mon/std from database	auto1.html/pdf [code][mn55na_sub] orig.-retrieved: mn55na-0pt.json auto-corrected: mn55na-1pt.json	AUTO[0] [0]: MONITOR, MONIT-REF, MONIT,..
2.	part3-2-user1	Correct data by user's program	Ratio	pu239/u235(n,f):(Ei)	user1.html/pdf [code][myCorr1] [pu239nf203sql][pu239nf203corr]	USER
3.	part3-3-expert1	Apply experts corrections stored in X4Pro database	SIG	Zn-64(n,p):CS(Ei)	expert1.html/pdf [code][corr_subr] [x13597002sub];terminal:output orig.-retrieved: orig.json corrected-by-expert-db: corr.json	EXPERT
4.	part3-4-auto2	Automatically renormalize CS using old monitor CS and new standards, old and new decay data	SIG	V-51(n,p):CS(Ei)	auto2.html/pdf [code][x12898sig_sub] orig.-retrieved: orig.json auto-corrected: auto.json	AUTO[0][1][2] [1]: DECAY-DATA [2]: DECAY-MON
5.	part3-5-ratio2sig	Retrieve ratios, calculate CS and uncertainties using latest standards	Ratio	V-51(n,p)/U-238(n,f):(Ei) Ratio→SIG	ratio2cs.html/pdf [code][x12898sig_sub] orig.-retrieved: ratio.json auto-corrected: sig.json	CALC: Ratio×U-238(n,f) → SIG
6.	part3-6-daleg2sig	Retrieve Legendre coefficient L[0], calculate cross sections and errors	LEG	B-11(n,el):DA,,LEG(Ei) L[0]→SIG	B11nel.html/pdf/json [code][rx4db2] terminal:output	CALC: L[0]×4π → SIG

7.	part3-7-legrs2da	Retrieve LEG/RS and SIG from the same Entry; calculate missing Legendre coefficient L_0 ; calculate angular distributions	LEG/RS SIG	$Cu-0(n,e):DA,,LEG/RS(Ei)$ $SIG \rightarrow L_0$ $L[0..n], \theta \rightarrow DA(\theta)$	Cu0nel_da.html/pdf/json0/json1 [code][rx5da][rx5legrs] terminal:output	FIND: SIG from <Entry, Reaction, Ei> CALC: $SIG/4\pi \rightarrow L_0$; CALC(Ei): $L[0..n], \theta \rightarrow DA(Ei, \theta)$
8.	part3-8-leg2r33	Retrieve Legendre coefficient calculate angular distributions store R33dat (draft of R33)	LEG	Li-6(p,he3):LEG(Ei)	leg2da.html/pdf/r33dat [code][rx4leg_ei][r33out]	CALC(Ei): $L[0..n], \theta \rightarrow DA(Ei, \theta)$
	Additional common modules:					11) [auto_corr.py] automatic correction of EXFOR data 12) [expert_corr.py] expert's correction from X4Pro

Part-4. Export contents of X4Pro to NoSQL databases (for NoSQL databases developers)

Python. Retrieve data from X4Pro and store them as JSON in NoSQL-Document database

**Note. Since 2022-08-30 X4Pro includes two tables presenting Subentries and Datasets in X5-json format.*

***Note. Since 2024-09-04 these two tables are included from X4Pro distribution on request only.*

****Note. X5-json is available on GitHub: <https://github.com/vzerkin/EXFOR-X5json>*

1.	part4-0-couchdb	Retrieve JSON presentation of Subent in X5-json format from Table x4pro_x4z, created a Dictionary "Entry" and store it as Document in CouchDB database	-NA-	JSON files for each Subent are retrieved row-by-row and collected into Python List	program: [code] Retrieved from X4Pro: --subent.001 10010001.json.txt --subent.002 10010002.json.txt stored: 10010.json in CouchDB --entries: list of documents in CouchDB --entry: 10010.json in CouchDB	[howto]
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Part-5. Search for mistakes in EXFOR library using X4Pro (for NRDC-EXFOR compilers)

python Retrieve data from X4Pro and store them as plain text, CSV and JSON files

sqlite3 Simple command-line program (shell) for SQLite

SQL Structured Query Language used as instructions for programs above.

1.	part5-0-sql 2024	1) NRDC-2024/A47: Analyse the zero values coded under the headings PARITY, ERR-T and DATA-ERR. 2) Text plotting DATA from a Dataset	-NA-	Find non-physical values of parity ($=0$) and uncertainties (<0)	program: [err-t.py] found: [csv] [json] program: [data-err.py] found: [csv] [json] program: [exec1sql.py] • inc.Energy $<=0$: [SQL] [CSV] [JSON] • Parity $=0$: [SQL] [CSV] [JSON] utility code: [sqlite3] script: [runme.sh] • inc.Energy $<=0$: [SQL] [-csv] [-box] • histogram: [SQL] result: [-header -box]	[howto]
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Part-6. X4Pro for evaluators: ratios to cross sections + automatic renormalization + evaluators flags and scores

1.	part6-0-eval1flags 2025	Summary of tables with evaluators' flags and scores	-NA-	Flags: accepted/rejected Scores: good/doubtful/outlier	Overview: [sql] [txt] Experts: [sql] [txt] IAEA-CRP review: [sql] [txt] Statistical test: [sql] [txt] ---Save eval-score tables to CSV/JSON: program: [exec2sql1.py] select: [sql] result: [csv] , [json] select: [sql] result: [csv] , [json]	[howto] EXFOR-online [evaluators_flags]
9.	part6-1-eval2sig 2025	Retrieve CS, ratios, monitor data; convert ratios to cross sections, renormalize CS to new standards and new decay data. MyPlot: advanced interactive graphics.	SIG	$(N, ABS), ALF \rightarrow SIG$ Ratio $\times std \rightarrow SIG$ $Zn-64(N, P): SIG \rightarrow renorm.$	Python: [code][x4evalscore][rx4dbr] • U235ng.html/pdf • Pu240nf.html/pdf • Al27na.html/pdf • Al27na2.html/pdf • zn64np.html/pdf MyPlot: [code] , [endf2plot_js.py] , [exfor2plot_js.py] • Pu240nf-myplot • Mn-55-myplot	CALC: $ALF \times SIG: (n, f) \rightarrow SIG(n, g)$ CALC: $(R1)/(R2) \times std(R2) \rightarrow SIG(R1)$ CALC: $SIG/mon \times std \rightarrow SIG$ MyPlot: plotting stored JSON files with more interactions (JavaScript)